

What Holds a Hand-Drawn Diagram?

Vasili Gavrilov 

Independent researcher

Abstract

Diagram editors ship an automatic layout; MySQL Workbench calls it **Arrange > Autolayout**. Underneath is a cost, a weighted sum of complaints: boxes overlap, connected boxes sit far apart, edge lengths vary. The program moves boxes until the sum stops falling. 4,890 hand drawings were just pooled into a corpus, and the standing proposal is to learn the cost's weights from drawings like these. That only works if a finished drawing sits at a minimum of the cost. Nobody had checked at the coordinates.

The check is simple. Take a diagram a person drew and accepted. Move each box by two per cent of the drawing width, in sixteen directions, and ask whether any move lowers a criterion; a box no move improves is held. We ran this on 853 working diagrams, biological pathways and business processes, on 136 lab drawings, and, as controls, on **neato**, **dot** and ELK layouts of the same graphs.

The answer is one-sided. Edge length and stress hold almost no box a person placed, while the same test holds **neato**'s own layouts fully, so it does see a minimum where one exists. Overlap is satisfied exactly. What does hold the drawings is a criterion the costs do not carry: alignment into rows and columns. It holds a fifth to most of the median hand diagram and almost nothing in any optimised, random or grid-snapped layout. A hand layout is a layered layout with partial alignment, and the 4,890-drawing corpus shows the same profile.

A cost fitted to hand coordinates therefore learns the trade-offs between its own terms, not the drawer's. Whether the person aligned the boxes or the editor's guide did, the coordinates cannot tell; for fitting it makes no difference, the term is held and missing either way. The fix is an addition: offer alignment among the terms, keep length and stress, whose preference evidence stands, and report how many boxes the fitted cost holds at layouts people accepted.

2012 ACM Subject Classification Human-centered computing → Graph drawings

Keywords and phrases graph drawing, layout aesthetics, inverse optimisation, alignment, empirical study

Category Track 2, Long Paper

Related Version A full version with every sensitivity inline is maintained beside this one.

Supplementary Material Code, parsed corpora and every table's script: <https://github.com/Anode1/cjitter>, directory `example/diagrams`.

Generative AI Declaration Author: Please use the `\GAIDecl` macro to state your Generative AI Declaration. If you did not use Generative AI, simply write: Generative AI was not used in the preparation of this article.

1 Introduction

A layout program places the boxes of a diagram by minimising an *energy*, a weighted sum of quantities the field has agreed are bad: edges that cross, boxes that overlap, edges of very unequal length, boxes that sit on edges they do not belong to [3, 2, 4, 5]. The weights are asserted by the tool's author and reach the user as one unparameterised command: MySQL Workbench documents **Arrange > Autolayout** in full as "Automatically arranges objects on the canvas", and Oracle SQL Developer Data Modeler's documented remedy for a bad result is Undo. Since 1994 the proposal has been to learn the weights instead, from layouts people made or preferred [15, 26, 14, 19, 28]. Such methods differ in what they need of their examples. Learning from preferences or selections [15, 26, 14, 28] needs only that an accepted



© Vasili Gavrilov;

licensed under Creative Commons License CC-BY 4.0

Leibniz International Proceedings in Informatics

LIPICs Schloss Dagstuhl – Leibniz-Zentrum für Informatik, Dagstuhl Publishing, Germany

XX:2 What Holds a Hand-Drawn Diagram?

layout rank well under the learned energy. Inverse optimisation, which fits an objective to solutions taken as optimal [22, 23, 24], needs the stronger property: that the coordinates themselves sit at a minimum. The stronger property has never been tested against human coordinates, no published layout method yet commits to it, and 4,890 hand-made drawings were just collected into one corpus [12], so the first coordinate fit is one download away. That property separates an energy describing where a person put a box from one describing how they ranked the result. Where it fails, the fitted weights are the criteria’s mutual trade-offs rather than the person’s, and the term the person did optimise stays unlearned because it was never offered.

The assumption has a direct test, one criterion at a time. Take a diagram a person drew. Move one box a small distance in each of several directions and ask whether any of the moves lowers the criterion. If none does, the box is *held*: the person has put it where the criterion wants it, or where the criterion has stopped caring. Do this for every box and the fraction held says how much of the layout is a local minimum of that criterion; the criterion’s value says how well it is met. A criterion that holds every box at value zero is one the person satisfied; one that holds no box is one the person did not consider, or traded away.

The test puts coordinates and judgement on opposite sides. People prefer low-stress drawings and read them better [11, 17], while the drawings people make carry 3 to 9 times the force-directed stress in those studies’ own tables [17, 16]; neither side ran a test at the coordinates. Here the coordinates people drew are held by no distance term at all in any corpus, while *neato*’s stress layouts are held at 1.00 under the same test. People prefer what they do not draw, with the caveat that preference and production are measured on different graphs and tasks; no study yet takes both from one corpus.

We ran the test on 853 diagrams of 15 to 40 boxes from three unrelated communities (Section 4), and on layouts of the same graphs by four tools. Section 6 carries the numbers; the findings are these.

1. Overlap holds every box of the median hand diagram at value zero in every corpus, as it does in every tool that removes overlaps; the term is nonzero in 4 to 8% of hand diagrams. People satisfy it exactly.
2. Length and stress hold a hand-placed box almost nowhere, median diagram 0.00 in all three corpora, while *neato*’s stress layouts are held at 1.00. No reweighting repairs it: a weight of 0.001 on either term drops the held fraction of the sum below 0.015.
3. Alignment into rows and columns holds 0.52, 0.21 and 0.87 against *neato*’s 0.06 and 0.07 for a grid-snapped random layout. The base energy omits it and the hand layouts carry it, by the person’s hand or the editor’s guide (Section 7), and BPMN’s lanes cannot impose it.
4. Reading direction does not close the gap. Flow holds 0.59 to 0.85 by hand where *dot* and *ELK* hold 1.00: the person tolerates a backward edge the layered tools remove.
5. Crossings and node–edge separation hold every box because they are flat at this radius, not because they are met.
6. The corpus of the standing proposal behaves the same: on its 1,072 drawings in the band, length and stress hold 0.00 and alignment 0.87.

The rest of the paper is the foundation behind the test (Section 2), the instrument (Section 3), the corpora, the method, the measurements, the alternatives not yet excluded (Section 7), and what is already known (Section 8).

2 Foundations

2.0.0.1 The energy.

A layout of a graph with N nodes is a point $x \in \mathbb{R}^{2N}$, two coordinates per node, the boxes' sizes held fixed. An aesthetic energy is

$$E_w(x) = \sum_{k=1}^K w_k T_k(x), \quad w_k \geq 0, \quad (1)$$

where each term T_k is a quantity to be made small. The tool's author asserts the w_k ; the field of multicriteria layout is the choice of the T_k and the w_k [2, 4].

2.0.0.2 Local minimum.

A point x is a local minimum of E if $E(y) \geq E(x)$ for every y within some distance r of x ; for differentiable E this implies $\nabla E(x) = 0$. Two terms here are not differentiable, crossings being a count and overlap having corners where boxes touch, so the test uses no gradient. Fix a radius d and a set V of unit directions in the plane (sixteen, equally spaced). For node i and direction v let $x + d v^{(i)}$ be the layout with node i moved by d along v and every other node fixed. Node i is *held* at radius d if

$$E(x + d v^{(i)}) \geq E(x) \quad \text{for every } v \in V, \quad (2)$$

and the estimand is

$$q_d(x) = \frac{1}{N} \# \{ i : \text{node } i \text{ is held at radius } d \}. \quad (3)$$

$q_d = 1$ says the layout is a local minimum of E over single-node moves of length d ; for smooth E and small d it says $\nabla E(x) = 0$ to within the resolution of V . Equation (2) compares two energies, so it is defined for every term and, evaluated per term, separates the criteria a person satisfied from those they did not. A capped descent, this study's first estimand, answers the same question for the sum only; it is retained as a secondary.

2.0.0.3 The inverse problem.

Given layouts $x^{(1)}, \dots, x^{(G)}$ that people accepted, inverse optimisation asks for the weights w under which they are nearest to stationary [23, 24]. With the directional differences $D_{iv} T_k = T_k(x + d v^{(i)}) - T_k(x)$ in hand, node i is held under w exactly when $\sum_k w_k D_{iv} T_k \geq 0$ for all v , a set of linear inequalities in w . The residual is the smallest total violation over w on the simplex, a linear program, and it is zero only if some weighting of the terms makes the layout a local minimum. A term with $D_{iv} T_k < 0$ for some v at almost every node (uniform edge length, at any layout whose edges are not all the same length) cannot take positive weight without violating the inequalities at those nodes, which is the zero weight the random-search fit found.

2.0.0.4 Step terms.

A count or a step function holds a node wherever it is flat, so its value is reported beside q_d (Appendix C).

113 2.0.0.5 The reference length.

114 The length and stress terms compare distances with a reference L . L fixed at the layout's
115 median edge length is in general not the scale at which the term is least for that layout,
116 so at a true minimum every node still has a scaling direction that lowers the term and the
117 converged reference is not held (neato's stress layouts: $q_{0.02} = 0.07$ to 0.12). L is therefore
118 the value at which the term is least for the layout as loaded, $\sum \ell^2 / \sum \ell$ over edges for length
119 and the same over $\|x_i - x_j\| / \delta_{ij}$ for stress, fixed before any move; neato's stress layouts are
120 then held at 1.00 in every corpus. The median and $1/\sqrt{N}$ are sensitivities.

121 The fit is Smelser, Miller and Kobourov's scale-normalised stress [9], Ahmed et al.'s for
122 the weighted form [10]; the pathology it repairs is theirs too (Appendix D).

123 2.0.0.6 The null and the controls.

124 A test that can only succeed is not a test. The *stationarity reference*: q_d on a layout a descent
125 has converged on must be near one, or the test lacks power for that energy; neato's layouts
126 are the reference for stress (1.00 in every corpus) and the uncapped climber's end point for
127 the sums (median 1.00 for every energy and corpus, Section 6). The *tool controls* of Section 5
128 say what a layout that does minimise something scores on each term. The *matched-budget*
129 *control*: uniform random placement at the same number of energy evaluations, which the
130 hand layout must beat, or "held under overlap" would be a property of sparse graphs and
131 not of the person.

132 3 The instrument

133 The descents, the control and the paired statistics come from **cjitter**, a C library of four
134 stochastic searches over a box of real variables: hill climbing with restarts, simulated annealing,
135 a genetic algorithm, and uniform sampling as the control. Every run is byte-reproducible
136 from its seed on every platform, and **cjitter_compare** runs all four at the same budget on
137 the same seeds and reports an exact paired sign test, Holm-corrected, against the control.
138 "Better than uniform sampling at equal cost" is the only claim it lets a search make.

139 Its **block** operator, one object moved a random step and the step kept when the summed
140 energy fell, is the operator of a label placer the author deployed in 2001; the directional test
141 of equation (2) is that proposal made exhaustive, every direction tried, none accepted, the
142 answer read off.

143 The library was written for a diagram task. MySQL Workbench's reverse-engineering
144 rebuilds a schema diagram auto-placed after every database change, and restoring a 44-table
145 diagram by hand cost the author about an hour each time, so **cjitter** places a migration's
146 new tables onto the frozen rest. Pinning the rest is justified only if the person's positions
147 carry something worth keeping; what they carry, at the coordinates, is this paper's question.

148 4 The corpora

149 Three collections of diagrams whose node coordinates a person chose, from three communities
150 with different authoring tools, all redistributable.

corpus	licence	files	parsed	15–40 nodes	used	median m/n
WikiPathways GPML, <i>Homo sapiens</i>	CC0	1,016	571	305	305	1.06
Reactome SBN-ML	CC0	1,382	710	248	248	1.04
BPMN Academic Initiative	academic	29,810	21,784	6,723	300	1.11

The layouts are working documents: Reactome diagrams are laid out by curators [32], WikiPathways pathways are drawn in PathVisio [33], the BPMN models are students' work in Signavio [34, 35]; nobody was asked to draw nicely, and the three tools share no layout code. Two WikiPathways files (WP5037, WP3391) carry coordinates converted from Reactome [36] and are dropped.

The BPMN archive is mixed: of the 21,784 models the parser reads, 13,435 have every edge a BPMN flow or association and the rest are EPC, Petri net and UML models, whose flow elements carry stencils outside the connector set and so become degree-2 nodes under the node rules. Only the 13,435 enter, 6,723 of them in the band, and the first 300 by model id are used; the two biological corpora are used whole.

A diagram enters as its largest connected component; container glyphs (SBGN compartments, BPMN pools and lanes) are not nodes; SBGN process glyphs are nodes, so the Reactome graphs are bipartite and their m/n is the process graph's; each layout is rescaled so that its bounding box fits the unit square, uniformly by the larger extent, so aspect is preserved, box sizes scaled with it. The directional test runs on 15 to 40 nodes; the climber pilot used 10 to 400. The registered deduplication pass finds one Reactome drawing stored under two ids and two structural near-pairs, none moving a median; clustering the WikiPathways interval by its 153 first authors widens it from [0.48, 0.54] to [0.48, 0.55] (Appendix J). The 41 to 100 node band is run in Section 6.

How the coordinates were made matters for the alignment criterion. Of the raw coordinates, 17% are integers in WikiPathways, 67% in Reactome and 63% in BPMN (44% multiples of five); the fraction of boxes sharing an exact x with another box is 0.37, 0.06 and 0.42, an exact y 0.29, 0.12 and 0.76, and either 0.54, 0.17 and 0.88, which is the figure the alignment row of Section 6 should be read against. In WikiPathways exact alignment is twice as common as integer coordinates, so it is not a rounding grid's act; the coordinates cannot say whether it is the person's hand or the editor's snap-to-object guide, which PathVisio draws on every drag. BPMN authors work on a grid inside lanes, hence the y figure of 0.76, and their editors guide every drag toward the neighbours; Reactome's curation tool rounds and its authors align little. Section 7 takes this up.

5 The method

5.0.0.1 Terms.

With n nodes, m edges, box i of width w_i and height h_i , L the reference length of Section 2, fitted to the layout under test:

The formulas, one line per term, are Appendix A; the comments in `energy.h` are normative.

The base energy is $T_C + T_O + T_L$ with weights (1, 1, 1), the form of the simulated-annealing layout of Davidson and Harel [2] and of the multicriteria tradition since [29]. The field's other default is stress [30, 31], $\sum_{i < j} (\|x_i - x_j\| - L \delta_{ij})^2 / \delta_{ij}^2$ with δ_{ij} the graph distance, which force-directed tools and `neato` minimise and which Dwyer et al. [17] found human choices track; it is the second base energy. T_R , T_A and T_N are the candidates for what the base energy misses. The alignment definition above is HOLA's gridiness [13], the published form, a step function of position. Three continuous forms are the sensitivities: A1, per node the distance to the nearest other node along whichever axis is nearer, which has a corner at exact alignment; A2, rows and columns priced separately; A3, a Gaussian kernel of width $s = 0.02$, smooth. The corner holds the box: A1 holds 0.52 of WikiPathways boxes and A3 holds 0.20.

XX:6 What Holds a Hand-Drawn Diagram?

200 5.0.0.2 Routes.

201 An edge is the polyline the file stores: the attachment point on the source box, the interior
202 waypoints, the attachment point on the target box; where the file stores nothing it is the
203 chord between the centres. Crossings, orthogonality and node-edge separation are evaluated
204 on the route; length and stress on the centres. When a node moves its attachment points
205 move with it and the waypoints stay. The stored-route statistics are Appendix D; the chords
206 are the sensitivity.

207 5.0.0.3 Reading direction.

208 T_F needs a direction u . It is one of the four axis directions, the one under which T_F is least
209 for the layout as loaded, fixed before any move, the same rule as for L . Undirected edge
210 kinds (GPML graphical lines, BPMN undirected associations) do not enter it.

211 5.0.0.4 Directional test.

212 Radius $d \in \{0.005, 0.01, 0.02, 0.05\}$ of the drawing width, sixteen directions, every node, each
213 term alone and the sums of the tables. Primary cell $d = 0.02$. The directions are exact to
214 double precision (a series evaluated in the first quadrant and rotated), and a move that lowers
215 the energy by less than 10^{-12} of its value is a tie; an earlier build with directions off by up
216 to 4×10^{-6} counted the shift that error made as an improvement and read alignment lower
217 by up to 0.04. Deterministic: no seed, no budget. Every median carries a 95% bootstrap
218 interval over diagrams (1000 resamples).

219 5.0.0.5 Tool controls.

220 The same graphs with the same box sizes and edge directions laid out by `neato` (`-Gstart=1`,
221 so the file reproduces), by `neato` with `-Goverlap=false` (prism), by `dot`, and by ELK
222 layered 0.12.0 (`elkjs`, direction RIGHT, every other option at its default), the engine the
223 BPMN and KIELER editors run; a check in the binary confirms each control is the same
224 graphs.

225 5.0.0.6 Descent, secondary.

226 Through the library: hill climbing from the hand layout, one node per proposal, each node
227 kept within a disc of radius d about its start, 4000 energy evaluations, 15 seeds; uniform
228 sampling in the same cap at the same budget on the same seeds, the matched-budget control;
229 the converged reference the same climber uncapped, 60,000 evaluations, q_d measured where
230 it stops. The step schedule, the reported columns and the pilot's step lesson are Appendix E.

231 5.0.0.7 Fitting.

232 Weights on the simplex, the linear program of Section 2 over every node and direction of
233 a corpus (about 120,000 inequalities, HiGHS), the residual its value per node; the residual
234 depends on the terms' scales and q does not, so both are reported, on the fitting diagrams
235 and held out under 5×2 cross-validation (five seeded halvings, each half fitted and scored on
236 the other). Inverse optimisation is degenerate whenever a term is satisfied exactly: overlap
237 holds every node, so any weight vector with all its weight on overlap has residual zero and
238 the program returns it. The informative output is the sweep, q as the weight on one term
239 grows from 0 with the rest on crossings and overlap.

5.0.0.8 Statistics.

The unit of inference is the diagram. Per corpus, hand against a control is the Wilcoxon signed-rank test with the exact sign test beside it and the Hodges–Lehmann shift with its distribution-free 95% interval, Holm-corrected within the (corpus, control) family; the mechanics and the script’s 630 self-checks are Appendix F. Equivalence claims (“no reweighting repairs it”) are read off the sweep, never off a failure to reject.

6 Measurements

The tables in this section are from one deterministic battery on the corpora of Section 4 (results/run_all.sh; commit, versions, seeds and corpus hashes in results/manifest.txt); the population, null and HOLA numbers ship with their scripts beside their committed outputs. The directional test is one deterministic pass over the whole band on the stored routes with the fitted L .

6.0.0.1 Each criterion alone.

$q_{0.02}$ on the hand layout, with its 95% bootstrap interval over diagrams, and on the three tool layouts of the same graphs; in parentheses the median value of the term at the layout, 0 when satisfied.

corpus		energy		hand	neato	prism	dot	elk
WikiPathways	crossings alone	1.00	[1.00, 1.00]	(0.000)	1.00 (0.000)	1.00 (0.000)	1.00 (0.029)	1.00 (0.036)
	overlap alone	1.00	[1.00, 1.00]	(0.000)	0.60 (0.030)	1.00 (0.000)	1.00 (0.000)	1.00 (0.000)
	length alone	0.00	[0.00, 0.00]	(0.219)	0.24 (0.011)	0.19 (0.014)	0.00 (0.241)	0.00 (0.127)
	stress alone	0.00	[0.00, 0.00]	(0.217)	1.00 (0.042)	0.94 (0.044)	0.00 (0.201)	0.00 (0.241)
	orthogonality alone	0.19	[0.18, 0.21]	(0.157)	0.00 (0.280)	0.00 (0.281)	0.19 (0.244)	0.19 (0.158)
	alignment A1 alone	0.52	[0.48, 0.54]	(0.003)	0.06 (0.011)	0.07 (0.010)	1.00 (0.000)	0.57 (0.001)
	alignment A3 alone	0.20	[0.18, 0.23]	(0.284)	0.03 (0.446)	0.03 (0.451)	0.45 (0.204)	0.35 (0.176)
	gridiness alone	0.87	[0.84, 0.89]	(0.424)	0.78 (1.000)	0.79 (1.000)	1.00 (0.125)	0.76 (0.200)
	node-edge alone	0.88	[0.86, 0.90]	(0.004)	0.89 (0.000)	1.00 (0.000)	0.78 (0.011)	0.74 (0.016)
	flow alone	0.59	[0.56, 0.62]	(0.131)	0.50 (0.211)	0.50 (0.211)	1.00 (0.000)	1.00 (0.000)
Reactome	crossings alone	1.00	[1.00, 1.00]	(0.000)	1.00 (0.000)	1.00 (0.000)	1.00 (0.056)	1.00 (0.066)
	overlap alone	1.00	[1.00, 1.00]	(0.000)	0.76 (0.022)	1.00 (0.000)	1.00 (0.000)	1.00 (0.000)
	length alone	0.00	[0.00, 0.00]	(0.234)	0.35 (0.005)	0.26 (0.009)	0.00 (0.232)	0.00 (0.205)
	stress alone	0.00	[0.00, 0.00]	(0.212)	1.00 (0.034)	0.85 (0.042)	0.00 (0.177)	0.00 (0.197)
	orthogonality alone	0.17	[0.15, 0.18]	(0.193)	0.00 (0.277)	0.00 (0.277)	0.15 (0.256)	0.12 (0.203)
	alignment A1 alone	0.21	[0.19, 0.24]	(0.004)	0.06 (0.009)	0.06 (0.010)	1.00 (0.000)	0.42 (0.002)
	alignment A3 alone	0.10	[0.08, 0.12]	(0.300)	0.03 (0.430)	0.03 (0.428)	0.39 (0.206)	0.31 (0.213)
	gridiness alone	0.65	[0.62, 0.68]	(0.640)	0.77 (1.000)	0.75 (1.000)	0.95 (0.176)	0.74 (0.306)
	node-edge alone	1.00	[0.94, 1.00]	(0.000)	0.93 (0.000)	1.00 (0.000)	0.80 (0.007)	0.75 (0.021)
	flow alone	0.62	[0.60, 0.67]	(0.096)	0.45 (0.217)	0.44 (0.222)	1.00 (0.000)	1.00 (0.000)
BPMN	crossings alone	1.00	[1.00, 1.00]	(0.000)	1.00 (0.000)	1.00 (0.000)	1.00 (0.000)	1.00 (0.000)
	overlap alone	1.00	[1.00, 1.00]	(0.000)	0.41 (0.073)	1.00 (0.000)	1.00 (0.000)	1.00 (0.000)
	length alone	0.00	[0.00, 0.00]	(0.291)	0.42 (0.006)	0.27 (0.014)	0.00 (0.250)	0.00 (0.273)
	stress alone	0.00	[0.00, 0.00]	(0.203)	1.00 (0.017)	0.90 (0.024)	0.00 (0.169)	0.00 (0.190)
	orthogonality alone	0.73	[0.70, 0.75]	(0.011)	0.00 (0.280)	0.00 (0.278)	0.26 (0.167)	0.25 (0.139)
	alignment A1 alone	0.91	[0.87, 0.97]	(0.000)	0.06 (0.011)	0.06 (0.011)	0.89 (0.000)	0.62 (0.001)
	alignment A3 alone	0.31	[0.29, 0.33]	(0.207)	0.00 (0.453)	0.00 (0.452)	0.19 (0.168)	0.20 (0.155)
	gridiness alone	0.95	[0.94, 0.97]	(0.176)	0.78 (1.000)	0.78 (1.000)	0.79 (0.219)	0.69 (0.250)
	node-edge alone	1.00	[1.00, 1.00]	(0.000)	1.00 (0.000)	1.00 (0.000)	0.82 (0.003)	0.76 (0.016)
	flow alone	0.85	[0.82, 0.87]	(0.045)	0.62 (0.121)	0.60 (0.123)	1.00 (0.000)	1.00 (0.000)

6.0.0.2 The sums.

The same test under the summed energies, equal weights; the grid is Appendix B. Overlap holds every box at value zero in every layout that removes overlaps; length and stress hold

almost no hand-placed box, 0.3 to 1.1 per cent per corpus, while **neato**'s stress layouts are held at 1.00, so the zero is the layout's and not the test's. Hand and **neato** part furthest on alignment A1; **dot** and ELK score high on it because a layer is a row, and a layout snapped to the same grid with no author holds 0.065 to 0.075 (Section 7). Under T_A , q counts boxes within $d/2$ of sharing a coordinate, so it tracks the exact-tie fractions of Section 4 to within a few points.

Crossings, overlap and node-edge are flat at this radius (their decrease and active columns: Appendix C), and their q is not read beside The three terms above the rule are the flat ones and their q should not be read beside the rest: on BPMN only 3% of diagrams offer a crossing-reducing move and 7% an the corpus's near-trees, not about the author. The five other terms are live everywhere, and the corpus's near-trees, not about the author. The five below are live everywhere, and there the 0.00 of length and stress is a real reading: every diagram has improving moves, the mean best one worth 1.2 to 1.3 per cent of the length term and 0.5 to 0.7 per cent of stress, and none of them was taken. One such move is below perception, so at any perceptual ϵ the layouts are ϵ -stationary under the distance terms; the failure is cumulative, not local, since from the same starts the capped climber removes 0.26 to 0.31 of $C+O+L$. A coordinate fit needs exact stationarity, and the layouts sit a quarter of the energy from it. On a displacement scale, a hand layout is as far from a stress minimum as a converged layout whose every box was moved by about the test radius (Appendix E).

The BPMN figure is the one number in the paper that the corpus's selection moves. The 300 models are the first by id of the 6,723 in the band, and on all 6,723 the alignment hold is 0.87 rather than 0.91, the mean 0.81; resampling puts the committed corpus at the 99th percentile of a random 300's [0.83, 0.90]. Every other term is where it was: crossings, overlap, node-edge and $C+O$ at 1.00, length, stress and $C+O+L$ at 0.00, orthogonality 0.733 against 0.726, A3 0.312 against 0.310 and flow 0.842 against 0.854 (**results/bpmn_population.md**). The tables here stay on the committed 300; a seeded random 300 with all four tool controls regenerated (**data/bpmnr.txt**, **results/bpmn_random300.md**) confirms the confinement: A1 0.846 around the population 0.872, everything else within 0.017. One pair changes verdict: on the random 300 hand reads -0.05 $[-0.10, -0.00]$ against **dot** on A1 (Holm $p = 0.042$), below **dot** as in the biological corpora; the committed sample's apparent parity was the selection's. The alignment claim is stated at 0.87, the population value. On the value the alignment comparison survives the step terms' flatness: 58, 36 and 82 per cent of hand boxes sit in an alignment of three or more, none of **neato**'s; the value-level reading of gridiness, node-edge, crossings and flow is Appendix C.

6.0.0.3 Sensitivities.

The radius, the direction count, the reference length, the box extent measured border to border, the tie tolerance and the routes closed to chords are each swept in Appendix D; the 1,008-cell grid over them flips no verdict, the A1 levels are sixteen-direction upper bounds, and the gap to every control survives at 64.

6.0.0.4 Descent and chance.

The secondary estimand's table is Appendix E; its load-bearing line is the power: the converged reference's median q is 1.00 for every energy and corpus, and the capped climber beats matched-budget uniform sampling on all 15 seeds, surviving Holm.

6.0.0.5 Reweighting.

The bracket beside each held-out q here is the range of the ten 5×2 values, a spread over correlated splits, not a confidence interval. The linear program is degenerate as Section 2 says it must be: on the two biological corpora it puts all weight on overlap (residual per node 0.0001 and 0.0000; held-out q 0.987 [0.982, 0.993] and 0.994 [0.990, 0.998]), on BPMN all weight on crossings (0.0009; 0.992 [0.979, 0.998]). Confined to length and stress it puts everything on stress and holds 0.003 to 0.011 of nodes: no distance-only weighting holds a hand layout. Confined to the four terms no layout satisfies exactly (orthogonality, alignment, node-edge, flow) it is identified, and what it identifies is the person’s term: alignment carries 0.95, 0.92 and 0.89 of the weight with the held-out q stable at 0.248 [0.236, 0.260], 0.149 [0.142, 0.157] and 0.660 [0.647, 0.674]. Offered all eight terms it returns to the degenerate vertex where one is open (WikiPathways’ held-out q swings [0.235, 0.991]); on BPMN, where no term is exact, it places 0.86 on alignment, stable at 0.514. An earlier draft ran this fit on the smooth A3 kernel and concluded that nothing is identified; that conclusion was the kernel’s, not the data’s. The sweep: the fraction of the corpus’s nodes held as a weight w on one term joins crossings and overlap at equal weight:

	WikiPathways		Reactome		BPMN	
term added to C+O	$w=0.001$	$w=1$	$w=0.001$	$w=1$	$w=0.001$	$w=1$
none ($w = 0$)	0.94		0.97		0.98	
length	0.008	0.004	0.010	0.007	0.012	0.009
stress	0.005	0.003	0.008	0.006	0.014	0.011
alignment A1	0.506	0.516	0.243	0.238	0.811	0.821
flow	0.576	0.606	0.613	0.625	0.797	0.810

The fall is a step, not a slope: whatever weight length or stress carries, 0.001 or 1, the sum holds under 0.015 of the nodes, because at almost every node some direction lowers the distance term and nothing else moves. Alignment and flow can carry weight: the criteria people partly satisfy survive in the sum at the fraction they are satisfied. Per diagram the same rows carry intervals: at $w = 0.001$ the median held fraction is 0.00 [0.00, 0.00] under either distance term and 0.52 [0.47, 0.54], 0.21 [0.19, 0.25] and 0.89 [0.86, 0.95] under A1 (`results/constrained.py`). Under the smooth A3 kernel the alignment row reads 0.29, 0.13 and 0.38 at $w = 0.001$: a differentiable proxy carries a third to half of what the corner carries, the number a gradient-trained consumer of this recommendation should expect.

6.0.0.6 Lanes.

82% of BPMN boxes sit in a lane or pool, and a horizontal lane constrains y , so the A1 hold of 0.91 could be the container’s (`results/lanes.py`). Per held box, the alignment that holds it: 30% a column, which no horizontal lane imposes; 13% a row shared with a box of another lane or of no lane, which no lane imposes either; 57% a row within one lane. Those lanes are a median 6.3 box heights tall (quartiles 3.7 and 9.4; 3% under two box heights), so the row placement inside them was chosen, not forced; where the box sits on its row, and the 43% held across or against the lanes, is the person’s.

6.0.0.7 The paired tests.

Per corpus, hand against `dot` on each criterion alone: the Hodges–Lehmann shift of $q_{0.02}$ with its 95% interval, zeros dropped, and the Holm-adjusted signed-rank p within the

XX:10 What Holds a Hand-Drawn Diagram?

corpus's family (`analyse.py`; the same families against `neato`, `prism` and ELK are in `results/tests/`); the table is Appendix F.

The shifts under crossings, overlap, length and stress are distinguishable but confined to the minority of diagrams where the pair differs at all. The large shifts are the profile's gaps: the hand layout sits below `dot` on alignment in the biological corpora (-0.46 and -0.70) and within $[-0.01, +0.09]$ of it on BPMN, a bound read off the interval, not an equivalence read off the p of 0.14 (the random 300 below reads -0.05); below it on flow everywhere (-0.17 to -0.40), and above it on node-edge separation and, in BPMN, on orthogonality ($+0.43$), the criteria `dot` trades away.

6.0.0.8 Reinsertion.

The constructive check: delete each box in turn and let an energy place it against the frozen rest (`reinsert.c`), the distance to the person's position the score. Every energy beats chance, 0.18 to 0.23 of the drawing width against 0.52 to 0.54, and adding A1 to $C+O+L$ moves the median distance not at all: the neighbourhood is the distance terms' choice. The rail is alignment's: with A1 offered, the box lands within 0.005 of the person's row or column in 0.127, 0.101 and 0.226 of placements against 0.085, 0.083 and 0.162 without it (chance 0.02, McNemar z up to 14.3; robustness in Appendix L).

6.0.0.9 The fourth corpus, restored.

The pre-registration named HOLA's formative drawings [13] and the previous draft dropped them without a word. Restored: 136 drawings, 17 participants each laying out the same 8 abstract graphs, scraped from the study's published SVGs (provenance beside `data/hola.txt`), chord edges, below this paper's 15-box band and free of every diagram notation. The pattern holds: overlap and crossings 1.00, length and stress 0.00, alignment 0.50, between Reactome's 0.21 and BPMN's 0.91; node-edge 1.00, so H4 fails here too (clustering and corpus agreement: Appendix J). These are drawings from a study tool rather than a diagram editor, which bears on the guide question of Section 7; the corpus does not record whether that tool drew alignment aids.

6.0.0.10 The motivating corpus.

The GD Collection itself [12] runs through the same test on the terms that need neither box extent nor edge direction: 1,072 of its 4,890 drawings have a largest component in the 15 to 40 band (chord edges, zero box sizes; the provenance and audit caveats stand, `results/gdcoll.py`). The profile is the paper's: length and stress hold 0.00, at exactly zero in 82 and 84% of drawings; crossings 1.00; orthogonality 0.06; alignment A1 holds 0.87 of the median drawing, with 0.43 of drawings at exactly 1.00 and 0.06 at zero. The corpus the standing proposal would fit first is held by the term the standard cost omits, at BPMN's level.

6.0.0.11 The 41 to 100 band.

The registered larger band, run against `neato` (`results/band41.md`; 108, 265 and 300 diagrams, the BPMN file the first of its band by id): every verdict stands. Length and stress hold 0.00 by hand in every corpus and `neato`'s stress holds 1.00; overlap 1.00; alignment 0.39, 0.16 and 0.74 against `neato`'s 0.07 to 0.08. The hold sits below the small band's 0.52, 0.21

and 0.91, so larger diagrams carry less alignment, and crossings stop holding every biological box (0.97 and 0.98): size buys improving crossing moves.

6.0.0.12 Null layouts.

Uniform random layouts of the same graphs put the chance line at 0.06 to 0.07 under alignment against hand 0.52, 0.21 and 0.91; length and stress hold 0.00 in the random layouts too, so the distance result separates optimisers from everyone, not humans from tools. A jitter of half a grid cell leaves alignment 0.76, 0.44 and 0.19 of its hold while orthogonality collapses, so orthogonality, not alignment, is the tie artefact. Axis-only moves still improve the distance terms, and the constrained model, stress subject to alignment [39, 40, 41], fails the same way: 0.87 to 0.90 of the median diagram improves without raising A1 (Appendix G).

6.0.0.13 The pre-registration's scorecard.

Four of the five registered hypotheses hold; H4, that alignment out-holds orthogonality and node-edge, is **refuted**, node-edge holding 0.88 to 1.00, and the omitted-term conclusion survives it because node-edge is priced in the standard energy and alignment is not (Appendix I; the pitch radius runs there too, and nothing registered is unrun).

7 What else could produce this

1. **The editor snapped to a grid**, so alignment is the tool's. Excluded. The declared half-pitch displacement cannot fail at these pitches, for a corner reason Appendix G derives; the control that does discriminate is a layout snapped to the same grid, same boxes, same bounding box and no author, and it holds 0.065, 0.075 and 0.067 against the hand 0.52, 0.21 and 0.91. Displacing the hand layouts past $d/2$ takes them to that level, and a random layout snapped to the plausible working pitch of five units holds 0.076, the same as at pitch one.
2. **The editor snapped to the neighbours**, so alignment is the guide's. Not separable on these corpora, and conceded. Snap-to-object guides escape any grid control, and the hold orders exactly as their affordances do: 0.87 to 0.91 where guides meet every drag, 0.52 where they are offered, 0.21 where they are absent. For fitting an energy the attribution changes nothing, the coordinates hold a term the energy lacks either way; for the word "optimise" it does, and the claim here is about the layouts, not the muscle. Separating them needs a guide-free editor or an interaction log, and these corpora carry neither. The nearest evidence is the restored HOLA corpus: drawings from a study tool, not a commercial editor, hold alignment at 0.50, so the phenomenon does not need Signavio to appear. The eras say the same (**results/era.md**): the GD Collection holds A1 at 0.81 in 1998 to 2005, when its figures came from xfig, Ipe and hand-coded PostScript with no shared guide, 0.60 in 2006 to 2014 and 0.95 in the TikZ years, where a typed coordinate is alignment chosen by hand; and WikiPathways is flat at 0.47 to 0.53 across four generations of WP numbers. No single editor's aid produces the hold; what stays open is only whether some aid does in each era.

The rest are excluded one by one in Appendix H, each with the test that did it and the numbers it turned on.

457 8 Related work

458 Aesthetic criteria and their weighting go back to Tamassia, Di Battista and Batini [3] and
 459 were made an explicit energy by Davidson and Harel [2]. Purchase [4] gave the criteria their
 460 metrics, and that list has no alignment metric. Whether people agree with the criteria has
 461 been asked three ways.

462 Three lines of evidence exist, tabulated with what each gives and leaves in Appendix M.
 463 By having people draw: van Ham and Rogowitz [16], Dwyer et al. [17], Purchase, Pilcher and
 464 Plimmer [25], HOLA's formative study [13], Mooney et al. [11], and the harvested collection
 465 [12]. By showing people drawings: [18, 14]. By learning from examples: [15, 26, 19, 28, 42]
 466 and O'Donovan, Agarwala and Hertzmann [8], who fit a design energy taking each example
 467 as optimal: the assumption examined here, and the closest prior instance of it we found.

468 Alignment as a human criterion is therefore known since 2012 in graph drawing, and since
 469 1985 in drawing editors: Pavlidis and Van Wyk's beautifier inferred alignment constraints
 470 from the near-alignments of a rough drawing [37], Igarashi et al. made the inference interactive
 471 [38], and the guides of Section 7 are their descendants. The new part is the count of how
 472 many hand-placed boxes it holds, on diagrams of 15 to 40 boxes, beside the counts for the
 473 criteria tools weight. Two results bound what such a count can mean: van Wageningen,
 474 Mchedlidze and Telea [20] show drawings with identical metric values that differ in quality,
 475 so a fitted energy with no residual would still not be the energy people use; the claims here
 476 stay on the negative side; and Kobourov, Pupyrev and Saket [27] found crossings matter
 477 little on sparse large graphs, consistent with the sparsity result.

478 The machinery is standard elsewhere. Inverse optimisation, the recovery of an objective
 479 from solutions taken as optimal, is due to Ahuja and Orlin [22] and Keshavarz, Wang and
 480 Boyd [23]; Chan, Lee and Terekhov [24] define the residual reported here. Matched-budget
 481 random controls are standard in hyperparameter search [6] and were prescribed as the
 482 baseline for genetic algorithms in 1995 [7]; we have not found them in layout. No paper
 483 fitting energy weights to human-placed coordinates was found in four searches of 2024 to
 484 2026 work.

485 9 Conclusion

486 Coordinates a person accepted are not minima of the energy in use, and the way they fail is
 487 specific: exact under overlap, nowhere near under uniform edge length or stress, and held
 488 instead by an alignment into rows and columns that, HOLA's gridiness step aside [13], no
 489 standard energy prices.

490 A method that learns weights from such layouts must offer alignment among its terms or
 491 it fits nothing the person did. For a tool builder that is one addition and one number: edge
 492 length and stress keep their weight, since a term inert at human coordinates is not thereby
 493 unwanted and the preference evidence runs the other way [11, 17]. The number to report is
 494 the fraction of boxes the energy holds when started from a layout a person accepted: $q_{0.02}$ as
 495 Section 5 fixes it, sixteen directions, fitted L , alignment A1, on layouts held out from any fit,
 496 with the decrease and active columns beside it, since the degenerate fit of Section 6 reaches
 497 0.99 by weighting a term everyone satisfies and a bare held fraction can always be bought
 498 that way. The charge is also bounded: dot and ELK hold alignment at 0.42 to 1.00 without
 499 pricing it, so the missing term convicts the force-directed and multicriteria energies, not the
 500 layered engines.

References

- 1 M. R. Garey, D. S. Johnson. Crossing Number is NP-Complete. *SIAM Journal on Algebraic and Discrete Methods* 4(3):312–316, 1983.
- 2 R. Davidson, D. Harel. Drawing Graphs Nicely Using Simulated Annealing. *ACM Transactions on Graphics* 15(4):301–331, 1996.
- 3 R. Tamassia, G. Di Battista, C. Batini. Automatic Graph Drawing and Readability of Diagrams. *IEEE Transactions on Systems, Man, and Cybernetics* 18(1):61–79, 1988.
- 4 H. C. Purchase. Metrics for Graph Drawing Aesthetics. *Journal of Visual Languages and Computing* 13(5):501–516, 2002.
- 5 S. G. Kobourov. Force-Directed Drawing Algorithms. In *Handbook of Graph Drawing and Visualization*, CRC Press, 2013.
- 6 J. Bergstra, Y. Bengio. Random Search for Hyper-Parameter Optimization. *Journal of Machine Learning Research* 13:281–305, 2012.
- 7 A. Juels, M. Wattenberg. Stochastic Hillclimbing as a Baseline Method for Evaluating Genetic Algorithms. *Advances in Neural Information Processing Systems 8* (NIPS 1995), MIT Press, 430–436, 1996.
- 8 P. O'Donovan, A. Agarwala, A. Hertzmann. Learning Layouts for Single-Page Graphic Designs. *IEEE Transactions on Visualization and Computer Graphics* 20(8):1200–1213, 2014.
- 9 K. Smelser, J. Miller, S. Kobourov. “Normalized Stress” is Not Normalized: How to Interpret Stress Correctly. *2024 IEEE Evaluation and Beyond: Methodological Approaches for Visualization* (BELIV), 41–50, 2024.
- 10 R. Ahmed, C. Erten, S. Kobourov, J. Lotz, J. Miller, H. Taraz. Size Should Not Matter: Evaluating Network Visualizations with Stress. *arXiv preprint arXiv:2408.04688*, 2024.
- 11 G. J. Mooney, J. Miller, M. Wybrow, S. Kobourov, H. C. Purchase. Stress in Graph Drawings: Perception, Preference, and Performance. In *33rd International Symposium on Graph Drawing and Network Visualization* (GD 2025), LIPIcs 357, 38:1–38:23, 2025.
- 12 G. J. Mooney, T. Hegemann, A. Wolff, M. Wybrow, H. C. Purchase. Universal Quality Metrics for Graph Drawings: Which Graphs Excite Us Most? In *33rd International Symposium on Graph Drawing and Network Visualization* (GD 2025), LIPIcs 357, 30:1–30:20, 2025.
- 13 S. Kieffer, T. Dwyer, K. Marriott, M. Wybrow. HOLA: Human-like Orthogonal Network Layout. *IEEE Transactions on Visualization and Computer Graphics* 22(1):349–358, 2016.
- 14 M. Klammler, T. Mchedlidze, A. Pak. Aesthetic Discrimination of Graph Layouts. *Journal of Graph Algorithms and Applications* 23(3):525–552, 2019.
- 15 T. Masui. Evolutionary Learning of Graph Layout Constraints from Examples. *Proc. UIST '94*, 103–108, 1994.
- 16 F. van Ham, B. Rogowitz. Perceptual Organization in User-Generated Graph Layouts. *IEEE Transactions on Visualization and Computer Graphics* 14(6):1333–1339, 2008.
- 17 T. Dwyer, B. Lee, D. Fisher, K. I. Quinn, P. Isenberg, G. Robertson, C. North. A Comparison of User-Generated and Automatic Graph Layouts. *IEEE Transactions on Visualization and Computer Graphics* 15(6):961–968, 2009.
- 18 H. C. Purchase, D. Archambault, S. Kobourov, M. Nöllenburg, S. Pupyrev, H.-Y. Wu. The Turing Test for Graph Drawing Algorithms. *Graph Drawing (GD 2020)*, LNCS 12590, 2020.
- 19 A. Bharadwaj, D. Gwizdala, Y. Kim, K. Luther, T. M. Murali. Flud: A Hybrid Crowd-Algorithm Approach for Visualizing Biological Networks. *ACM Transactions on Computer-Human Interaction* 29(1), 2022.
- 20 S. van Wagoningen, T. Mchedlidze, A. Telea. Same Quality Metrics, Different Graph Drawings. *Graph Drawing (GD 2025)*, arXiv:2508.15557, 2025.
- 21 G. Di Battista, A. Garg, G. Liotta, R. Tamassia, E. Tassinari, F. Vargiu. An Experimental Comparison of Four Graph Drawing Algorithms. *Computational Geometry* 7(5–6):303–325, 1997.
- 22 R. K. Ahuja, J. B. Orlin. Inverse Optimization. *Operations Research* 49(5):771–783, 2001.

XX:14 What Holds a Hand-Drawn Diagram?

- 552 23 A. Keshavarz, Y. Wang, S. Boyd. Imputing a Convex Objective Function. *IEEE International Symposium on Intelligent Control*, 613–619, 2011.
- 553
- 554 24 T. C. Y. Chan, T. Lee, D. Terekhov. Inverse Optimization: Closed-Form Solutions, Geometry, and Goodness of Fit. *Management Science* 65(3):1115–1135, 2019.
- 555
- 556 25 H. C. Purchase, C. Pilcher, B. Plimmer. Graph Drawing Aesthetics: Created by Users, Not Algorithms. *IEEE Transactions on Visualization and Computer Graphics* 18(1):81–92, 2012.
- 557
- 558 26 M. Spönemann, B. Duderstadt, R. von Hanxleden. Evolutionary Meta Layout of Graphs. *Diagrams 2014*, LNCS, 16–30, 2014.
- 559
- 560 27 S. G. Kobourov, S. Pupyrev, B. Saket. Are Crossings Important for Drawing Large Graphs? *Graph Drawing (GD 2014)*, LNCS, 234–245, 2014.
- 561
- 562 28 P. Zhang, X. Li, X. Wang, H.-W. Shen, Y. Hu. Beauty in the Eye of AI: Aligning LLMs and Vision Models with Human Aesthetics in Network Visualization. arXiv:2604.03417, 2026.
- 563
- 564 29 R. Ahmed, F. De Luca, S. Devkota, S. Kobourov, M. Li. Graph Drawing via Gradient Descent, $(GD)^2$. *Graph Drawing (GD 2020)*, LNCS, 3–17, 2020.
- 565
- 566 30 T. Kamada, T. Kawai. An Algorithm for Drawing General Undirected Graphs. *Information Processing Letters* 31:7–15, 1989.
- 567
- 568 31 E. R. Gansner, Y. Koren, S. C. North. Graph Drawing by Stress Majorization. *Graph Drawing (GD 2004)*, LNCS, 239–250, 2005.
- 569
- 570 32 A. Fabregat et al. Reactome Diagram Viewer: Data Structures and Strategies to Boost Performance. *Bioinformatics* 34(7):1208–1214, 2018.
- 571
- 572 33 A. Agrawal, H. Balci, H. Hanspers et al. WikiPathways 2024: Next Generation Pathway Database. *Nucleic Acids Research* 52(D1):D679–D689, 2024.
- 573
- 574 34 M. Kunze, A. Luebbe, M. Weidlich, M. Weske. Towards Understanding Process Modeling: The Case of the BPM Academic Initiative. *BPMN 2011*, LNBIP, 44–58, 2011.
- 575
- 576 35 D. Sola, C. Warmuth, B. Schäfer, P. Badakhshan, J.-R. Rehse, T. Kampik. SAP Signavio Academic Models: A Large Process Model Dataset. *Process Mining Workshops (ICPM 2022)*, LNBIP, 453–465, 2023.
- 577
- 578
- 579 36 A. Bohler et al. Reactome from a WikiPathways Perspective. *PLOS Computational Biology* 12(5):e1004941, 2016.
- 580
- 581 37 T. Pavlidis, C. J. Van Wyk. An Automatic Beautifier for Drawings and Illustrations. *SIG-GRAPH '85*, 225–234, 1985.
- 582
- 583 38 T. Igarashi, S. Matsuoka, S. Kawachiya, H. Tanaka. Interactive Beautification: A Technique for Rapid Geometric Design. *UIST '97*, 105–114, 1997.
- 584
- 585 39 T. Dwyer, Y. Koren, K. Marriott. IPSep-CoLa: An Incremental Procedure for Separation Constraint Layout of Graphs. *IEEE Transactions on Visualization and Computer Graphics* 12(5):821–828, 2006.
- 586
- 587
- 588 40 T. Dwyer, K. Marriott, M. Wybrow. Dunnart: A Constraint-Based Network Diagram Authoring Tool. *Graph Drawing (GD 2008)*, LNCS 5417, 420–431, 2009.
- 589
- 590 41 J. Hoffswell, A. Borning, J. Heer. SetCoLa: High-Level Constraints for Graph Layout. *Computer Graphics Forum* 37(3):537–548, 2018.
- 591
- 592 42 X. Wang, K. Yen, Y. Hu, H.-W. Shen. DeepGD: A Deep Learning Framework for Graph Drawing Using GNN. *IEEE Computer Graphics and Applications* 41(5):32–44, 2021.
- 593

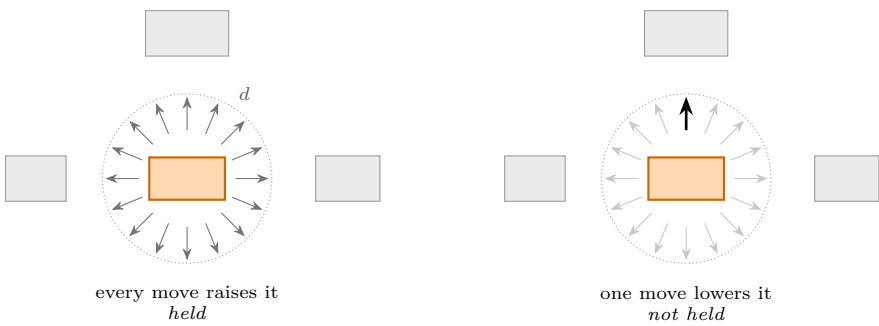


Figure 1 The test, one box and one criterion. The box is moved a distance d , two per cent of the drawing’s width, in each of sixteen exact directions, and the criterion is recomputed. A box is held when no move lowers it.

A The terms in full

term	definition	zero when
T_C crossings	proper crossings between segments of two routes with no shared endpoint, $/m$	planar
T_O overlap	overlap area of boxes / total box area	no overlap
T_L length uniformity	mean over edges of $((\ell - L)/L)^2$	all edges of length L
T_R orthogonality	mean over edges of the length-weighted mean over its segments of $\min(\Delta x , \Delta y)/(\Delta x + \Delta y)$	routes axis-aligned
T_A alignment	1 – fraction of nodes in an alignment of ≥ 3 (tol. 0.005)	every node aligned
T_N node–edge	squared shortfall of node to foreign route distance below $(w + h)/4$	none within
T_F flow	mean over directed edges of the backward component along the reading direction u , over length	no edge runs back

B The sums under composite energies

At the hand layout the three base terms share the energy as length 0.99, crossings 0.01, overlap 0.00, and $C + O + L$ inherits the length term’s zero.

XX:16 What Holds a Hand-Drawn Diagram?

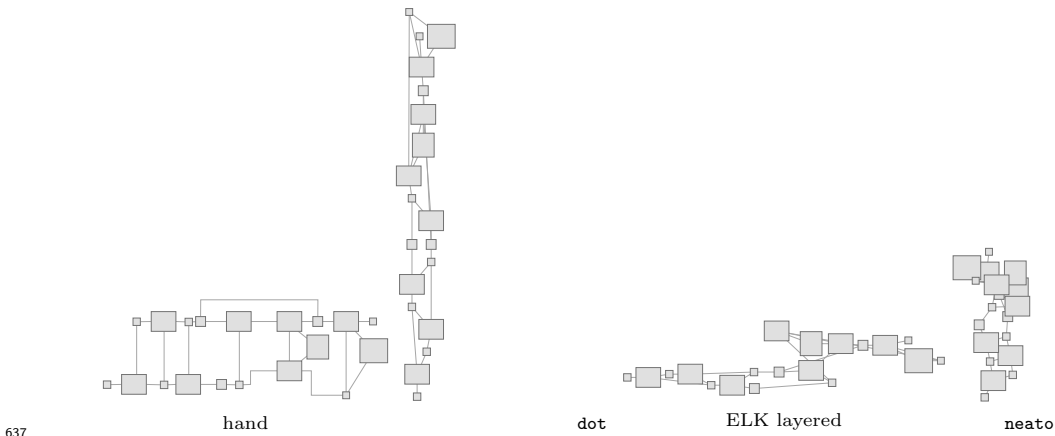


Figure 2 The same graph, the same box sizes, four layouts, one scale. The hand drawing and the two layered tools share the profile the test reads: held under overlap, free under length and stress. **neato** minimises stress and is the control that gives the test its power, held at 1.00 under stress where every hand layout reads 0.00. The tool layouts carry no waypoints, so their edges are drawn as chords.

609

	corpus	energy		hand	neato	prism	dot	elk
610	WikiPathways	C+O	1.00	[1.00, 1.00]	0.60	1.00	1.00	1.00
611		C+O+L	0.00	[0.00, 0.00]	0.17	0.19	0.00	0.00
612		C+O+S	0.00	[0.00, 0.00]	0.60	0.94	0.00	0.00
613		C+O+R	0.21	[0.19, 0.23]	0.00	0.00	0.19	0.22
614		C+O+A1	0.51	[0.47, 0.53]	0.04	0.07	1.00	0.57
615		C+O+A3	0.24	[0.22, 0.27]	0.03	0.03	0.50	0.41
616		C+O+grid	0.84	[0.81, 0.87]	0.45	0.79	0.96	0.74
617		C+O+N	0.86	[0.83, 0.88]	0.55	1.00	0.78	0.74
618		C+O+F	0.56	[0.53, 0.59]	0.30	0.50	1.00	1.00
619	Reactome	C+O	1.00	[1.00, 1.00]	0.76	1.00	1.00	1.00
620		C+O+L	0.00	[0.00, 0.00]	0.33	0.27	0.00	0.00
621		C+O+S	0.00	[0.00, 0.00]	0.76	0.89	0.00	0.00
622		C+O+R	0.17	[0.16, 0.19]	0.00	0.00	0.14	0.13
623		C+O+A1	0.22	[0.19, 0.25]	0.05	0.07	1.00	0.43
624		C+O+A3	0.11	[0.09, 0.12]	0.03	0.03	0.42	0.33
625		C+O+grid	0.64	[0.61, 0.67]	0.52	0.75	0.94	0.73
626		C+O+N	0.94	[0.91, 0.95]	0.71	1.00	0.80	0.75
627		C+O+F	0.61	[0.58, 0.65]	0.33	0.44	1.00	1.00
628	BPMN	C+O	1.00	[1.00, 1.00]	0.41	1.00	1.00	1.00
629		C+O+L	0.00	[0.00, 0.00]	0.24	0.29	0.00	0.00
630		C+O+S	0.00	[0.00, 0.00]	0.43	0.92	0.00	0.00
631		C+O+R	0.73	[0.71, 0.75]	0.00	0.00	0.28	0.40
632		C+O+A1	0.89	[0.86, 0.95]	0.04	0.09	0.88	0.68
633		C+O+A3	0.33	[0.30, 0.37]	0.03	0.03	0.33	0.45
634		C+O+grid	0.95	[0.93, 0.96]	0.28	0.78	0.75	0.67
635		C+O+N	1.00	[1.00, 1.00]	0.40	1.00	0.82	0.76
636		C+O+F	0.84	[0.81, 0.87]	0.26	0.61	1.00	1.00

C Decrease and active, term by term

The gridiness value puts 58%, 36% and 82% of hand-placed boxes in an alignment of three or more, none of **neato**'s, 88%, 82% and 78% of **dot**'s and 80%, 69% and 75% of **ELK**'s. The gridiness q of 0.77 to 0.78 on **neato**'s layouts, at value 1.000, is the step term's flatness, so alignment is compared on the value. Node-edge separation is nearly satisfied in every layout

(0.000 to 0.021) and holds 0.74 to 1.00 for the same reason. Crossings hold every box at this radius; the median value on the routes is 0.000 by hand in every corpus and 0.000 for **neato**, 0.029, 0.056 and 0.000 for **dot**, 0.036, 0.066 and 0.000 for ELK. Under flow **dot** and ELK are held at 1.00 with the term at 0.000 in every corpus; the hand layouts hold 0.59, 0.62 and 0.85 at 0.131, 0.096 and 0.045, and **neato**, which knows no direction, 0.44 to 0.62 at 0.121 to 0.222.

C.0.0.1 Step terms.

A count (crossings) or a step function of position (gridiness, Section 5) is flat almost everywhere and holds a node wherever no move of length d changes it. There q_d measures how much of the layout the criterion has stopped acting on, not how well it is met, so the term's value is reported beside q_d .

An interval of [0.00, 0.00] or [1.00, 1.00] is a pileup at the boundary, not precision: 87 to 92% of hand diagrams sit at exactly 0.00 under length and stress, and 62 to 85% of **neato**'s at exactly 1.00 under stress (**results/robustness.md**).

C.0.0.2 What q counts, term by term.

A held fraction is not commensurable across terms of different shape, and reading down a column compares three different things. Beside each q the table gives *dec*, the median over diagrams of the per-diagram mean best decrease a single move finds as a fraction of the term's value, and *active*, the share of diagrams in which any box has an improving move at all. Where *active* is small the term is flat at this radius: it holds every box without being satisfied by any of them, and its 1.00 carries no information about the person.

	WikiPathways			Reactome			BPMN		
term	q	dec	active	q	dec	active	q	dec	active
crossings	1.00	0.000	30%	1.00	0.000	20%	1.00	0.000	3%
overlap	1.00	0.000	18%	1.00	0.000	4%	1.00	0.000	7%
node-edge	0.88	0.047	64%	1.00	0.000	49%	1.00	0.000	15%
length	0.00	0.013	100%	0.00	0.013	100%	0.00	0.012	100%
stress	0.00	0.005	100%	0.00	0.006	100%	0.00	0.007	100%
orthogonality	0.19	0.012	100%	0.17	0.027	100%	0.72	0.008	99%
alignment A1	0.52	0.066	92%	0.21	0.069	99%	0.91	0.032	56%
flow	0.59	0.010	90%	0.62	0.012	95%	0.85	0.005	74%

D Sensitivities

The attachment point matters: of the 2,584 BPMN routes that carry an interior waypoint, 0.27 are axis-aligned at every segment when the route is closed at the box centres and 0.88 when it is closed at the stored attachments.

This is not a new device. Minimising a stress term over a scale factor is Smelser, Miller and Kobourov's scale-normalised stress [9], who give the minimiser in closed form and read it as a correlation coefficient, and Ahmed et al. [10] derive the same for the δ_{ij} -weighted graph stress used here. Their result is also why the fixed- L reading above is not a curiosity: under scale-sensitive stress they find a random layout ranked ahead of both **neato** and SFDP on nearly every graph of two 1,000-graph benchmarks. A control that reads 0.07 at a fixed L is that pathology, and fitting L is the known repair.

XX:18 What Holds a Hand-Drawn Diagram?

690 D.0.0.1 Radius.

691 q_d at $d \in \{0.005, 0.01, 0.02, 0.05\}$, hand layouts, then **neato**'s (`profile.py -sweep`):

692	WikiPathways				Reactome				BPMN			
693 energy	.005	.01	.02	.05	.005	.01	.02	.05	.005	.01	.02	.05
694 overlap, hand	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
695 length, hand	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
696 stress, hand	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
697 alignment A1, hand	0.52	0.54	0.52	0.44	0.12	0.17	0.21	0.19	0.90	0.91	0.91	0.87
698 gridiness, hand	0.94	0.91	0.87	0.75	0.82	0.75	0.65	0.57	1.00	1.00	0.95	0.88
699 flow, hand	0.59	0.59	0.59	0.59	0.62	0.62	0.62	0.63	0.85	0.85	0.85	0.85
700 overlap, neato	0.60	0.60	0.60	0.60	0.75	0.76	0.76	0.76	0.41	0.41	0.41	0.43
701 length, neato	0.04	0.12	0.24	0.47	0.07	0.19	0.35	0.58	0.08	0.22	0.42	0.72
702 stress, neato	0.90	0.96	1.00	1.00	0.96	1.00	1.00	1.00	0.94	0.97	1.00	1.00
703 alignment A1, neato	0.04	0.05	0.06	0.06	0.03	0.05	0.06	0.07	0.03	0.05	0.06	0.07
704 gridiness, neato	0.93	0.86	0.78	0.65	0.90	0.83	0.77	0.61	0.92	0.85	0.78	0.67
705 flow, neato	0.50	0.50	0.50	0.50	0.45	0.45	0.45	0.45	0.62	0.62	0.62	0.62

706 The hand verdicts do not move with the radius. **neato**'s under length do: a smooth term
707 near a minimum with gradient g and curvature λ holds a node when $|g| < d\lambda/2$, so a layout
708 that is nearly a minimum is held at a large radius and free at a small one, and **neato**'s
709 layouts, which minimise stress and not length uniformity, read as nearly uniform at $d = 0.05$
710 (0.47 to 0.70) and not at $d = 0.005$ (0.04 to 0.08). A hand layout at 0.00 across the range is
711 far from the minimum at every scale the test resolves. Gridiness falls with d in every layout
712 because a larger move reaches an alignment from further away; flow, a projection, does not
713 move with d at all. ELK's block (`results/sweep.md`) reads as dot's: length and stress 0.00
714 at every radius, flow 1.00 at every radius.

715 D.0.0.2 Directions.

716 Each doubling of V contains the previous set, so q can only fall as directions are ad-
717 ded and the primary sixteen-direction figures are upper bounds on the continuum value
718 (`results/robustness.md`). At 8, 16, 32 and 64 directions the verdicts stand: hand length
719 and stress 0.00 in every corpus at every count, **neato**'s stress 1.00 at every count. The
720 alignment level is what moves: A1 by hand reads 0.58, 0.52, 0.47, 0.39 on WikiPathways,
721 0.31, 0.21, 0.14, 0.07 on Reactome and 0.92, 0.91, 0.90, 0.86 on BPMN across the four
722 counts, while **neato**'s 0.06 falls to 0.00. The hand-to-control gap survives at 64; the levels
723 are comparable only at one count and are stated at sixteen. The full grid of radius, direction
724 count, reference-length rule and route treatment, 1,008 cells, flips no verdict: under the fitted
725 L the hand distance terms read exactly 0.00 in every cell (`results/specgrid.md`). The tie
726 tolerance, the last instrument constant, moves nothing at all: the A1 medians are identical
727 at 0.001 through 0.01 (`results/tolsweep.md`), so the alignments the test counts are exact
728 ties, not tolerance-window artefacts.

729 D.0.0.3 Reference length.

730 Under the median edge length instead of the fitted L , **neato**'s stress layouts score 0.07, 0.08
731 and 0.12 under stress and the hand layouts 0.00; under length the hand layouts score 0.03,
732 0.04 and 0.05 and **neato**'s 0.26, 0.41 and 0.45. Under $L = 1/\sqrt{N}$ no box of any layout is
733 held under either term, the scale being wrong for every layout. The hand verdicts do not
734 depend on the choice; the control's does, so the fit is primary (Section 2).

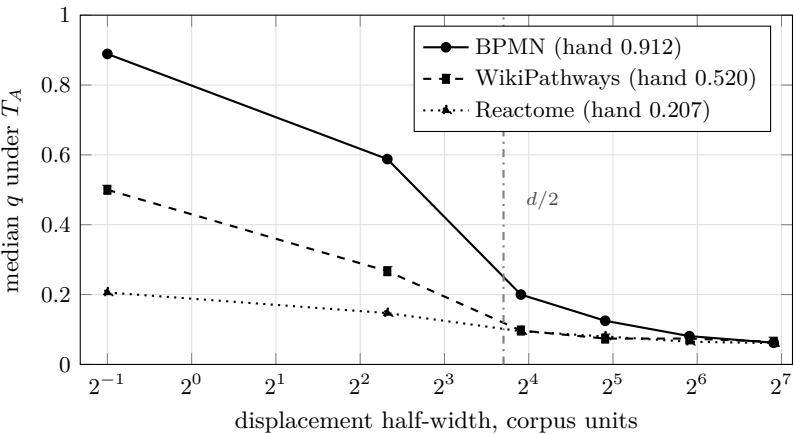


Figure 3 The jitter sweep. The declared half-pitch test sits at the left edge, 0.5 units, where nothing has changed. The corner of T_A protects a single displaced box below $\epsilon = d/2$, but every box is displaced, relative offsets reach 2ϵ , and the hold falls well before $d/2$: BPMN 0.89 to 0.59 by a half-width of 5. By $d/2$ every corpus is at the grid-snapped level of 0.06 to 0.08.

D.0.0.4 Box extent.

Centre distances on box diagrams could be the objection: with large, uneven boxes a person could sit at a minimum of a border-to-border distance term while failing the centre one. Measured border to border, each box’s offset to its border along the joining direction subtracted and L refitted on those distances, the hand verdicts do not move: length and stress hold 0.00 in every corpus, 0.86 to 0.92 of diagrams at exactly zero (`results/border.py`). `neato`, which minimised the centre form, is not held under the border form (0.00 to 0.09), as a layout optimising one criterion and tested on another should not be.

D.0.0.5 Routes.

The tables of Section 6 are on the stored routes; the chords are the sensitivity, `chords.md` beside the tables in the repository. Closing every edge at the box centres moves no verdict: crossings still hold every box (the WikiPathways value rises from 0.000 to 0.038, one crossing per 26 edges, the anchors’ work), orthogonality falls from 0.19, 0.17 and 0.73 to 0.16, 0.05 and 0.40, and node–edge separation from 0.88, 1.00 and 1.00 to 0.76, 0.85 and 0.87. The drawn geometry only raises the hand layout’s standing on the criteria it touches.

Each format’s stored routes, as a fraction of the edges in the band:

corpus	edges	with a waypoint	by what the file stores
WikiPathways	8,253	0.291	chord 0.587, anchor 0.225, elbow 0.154, curved 0.026, segmented 0.008
Reactome	6,827	0.069	chord 0.931, stored 0.069
BPMN	7,910	0.327	chord 0.673, stored 0.327

GPML’s Elbow and Curved connectors store their endpoints only and PathVisio computes the bends at draw time, so those routes read as chords; a quarter of WikiPathways edges end on an anchor of another interaction, where the person drew no segment to the node and the last segment is manufactured. Only the BPMN routes are orthogonal: of the WikiPathways and Reactome routes with a waypoint, 0.009 and 0.007 are axis-aligned at every segment, the biological formats drawing diagonal segments. The chords are the sensitivity.

765 **E** Descent, chance and the displacement yardstick

766 **E.0.0.1 The descent, in full.**

767 Through the library: hill climbing from the hand layout, one node per proposal, first step
 768 $d/2$ per coordinate, the step halved after 200 rejected proposals, each node kept within a disc
 769 of radius d about its start by the repair, 4000 energy evaluations, 15 seeds per instance; and
 770 uniform sampling in the same cap at the same budget on the same seeds, the matched-budget
 771 control. Reported per instance: the fraction of the cap used (mean over nodes of the distance
 772 moved over d), the fraction of energy removed, for the climber and for the control, and the
 773 number of seeds on which the climber ends lower than the control with its exact sign test.
 774 The climber pilot used a step of 0.1, five times the cap, so that nearly every proposal landed
 775 on the cap circle; the base-energy result did not depend on it (98.5 to 95.9% of the cap used
 776 as the step fell from 0.1 to 0.005) and the alignment result did (47 to 18%), so the directional
 777 test is primary. The converged reference is the same climber uncapped, 60,000 evaluations,
 778 with q_d measured where it stops.

779 **E.0.0.2 Descent and chance.**

780 The secondary estimand, per corpus and energy: the median fraction of the cap the capped
 781 climber uses and of the energy it removes, the same for uniform sampling in the cap at the
 782 same budget and seeds, the share of diagrams where the climber ends lower on at least 12 of
 783 15 seeds, and $q_{0.02}$ where the uncapped climber stops, the converged reference.

784 corpus	785 energy	cap used		energy removed		$\geq 12/15$	q conv.
		climb	random	climb	random		
786 WikiPathways	C+O+L	0.99	0.76	0.28	0.16	0.88	1.00
	C+O+A1	0.39	0.74	0.71	0.17	0.75	1.00
	stress	1.00	0.77	0.12	0.05	1.00	1.00
	C+O+F	0.44	0.76	0.27	0.18	0.77	1.00
790 Reactome	C+O+L	1.00	0.76	0.31	0.16	0.93	1.00
	C+O+A1	0.56	0.74	0.99	0.37	0.80	1.00
	stress	1.00	0.77	0.13	0.06	1.00	1.00
	C+O+F	0.36	0.75	0.32	0.22	0.71	1.00
794 BPMN	C+O+L	1.00	0.77	0.26	0.12	0.99	1.00
	C+O+A1	0.11	0.73	0.24	-0.01	0.95	1.00
	stress	1.00	0.77	0.14	0.07	1.00	1.00
	C+O+F	0.16	0.74	0.09	0.04	0.66	1.00

798 Under the distance energies the climber walks every node to the cap circle (0.99 to 1.00
 799 of d) and is still removing energy when the budget ends; under C+O+A1 it moves 0.11 to
 800 0.56 of the cap, the misalignments being nearer than d . The climber beats the control on all
 801 15 seeds, which survives Holm over the corpus (exact sign $p = 3.1 \times 10^{-5}$ against 0.05/305),
 802 on 0.81 to 0.99 of diagrams under C+O+L and on all of them under stress. The converged
 803 reference's median q is 1.00 for every energy and corpus, the test's power on the sums; the
 804 fraction of diagrams whose reference stops below 0.9 is at most 0.02 everywhere but C+O+F
 805 (0.05, 0.07 and 0.13), the flow term's plateaus.

806 **E.0.0.3 The displacement yardstick.**

807 Perturb `neato`'s converged stress layouts, every box moved a known fraction of the width in
 808 a seeded direction, and run the same capped climber (`results/calib.py`). Unperturbed it
 809 removes 0.001 of the stress energy; at a displacement of 0.005, 0.007 to 0.024; at 0.02, 0.09

to 0.26; at 0.05, 0.24 to 0.43; at 0.10 less again, the cap seeing a shrinking share of a growing energy. The hand layouts' 0.12 to 0.14 under stress sits at a displacement of 0.014 to 0.025.

F The paired tests in full

F.0.0.1 The statistics, in full.

The unit of inference is the diagram. Per corpus, the paired comparison of q_d between the hand layout and a tool control is the Wilcoxon signed-rank test, zeros dropped, ties by midranks, exact on the observed ranks up to 50 non-zero pairs and the normal approximation with continuity and tie corrections above, with the exact sign test beside it and the Hodges–Lehmann shift with its distribution-free 95% interval; the family is the terms compared against one control within one corpus, Holm-corrected. The script carries 630 self-checks against enumeration and against R. Equivalence claims (“no reweighting repairs it”) are read off the sweep, never off a failure to reject.

The columns are those of Section 6: the Hodges–Lehmann shift of $q_{0.02}$, hand against dot, with the Holm-adjusted signed-rank p .

824 energy	WikiPathways				Reactome				BPMN			
	HL [95%]		p		HL [95%]		p		HL [95%]		p	
crossings	−0.04	[−0.07, −0.01]	0.016	+0.02	[−0.01, +0.05]	0.30	+0.10	[+0.08, +0.12]	<1e-4			
overlap	−0.14	[−0.18, −0.11]	<1e-4	−0.07	[−0.10, −0.06]	0.0024	−0.22	[−0.29, −0.11]	<1e-4			
length	−0.01	[−0.03, +0.00]	0.27	−0.03	[−0.05, −0.01]	0.017	−0.09	[−0.12, −0.06]	<1e-4			
stress	−0.03	[−0.04, −0.00]	0.022	−0.03	[−0.05, −0.01]	0.0048	−0.08	[−0.11, −0.05]	<1e-4			
orthogonality	+0.01	[−0.00, +0.04]	0.27	+0.02	[+0.00, +0.04]	0.036	+0.43	[+0.41, +0.45]	<1e-4			
alignment A1	−0.46	[−0.49, −0.43]	<1e-4	−0.70	[−0.73, −0.67]	<1e-4	+0.04	[−0.01, +0.09]	0.14			
alignment A3	−0.25	[−0.28, −0.22]	<1e-4	−0.28	[−0.31, −0.26]	<1e-4	+0.13	[+0.10, +0.16]	<1e-4			
gridiness	−0.13	[−0.15, −0.11]	<1e-4	−0.29	[−0.32, −0.26]	<1e-4	+0.21	[+0.17, +0.25]	<1e-4			
node-edge	+0.13	[+0.10, +0.16]	<1e-4	+0.18	[+0.15, +0.21]	<1e-4	+0.26	[+0.23, +0.28]	<1e-4			
flows	−0.40	[−0.42, −0.38]	<1e-4	−0.33	[−0.35, −0.31]	<1e-4	−0.17	[−0.19, −0.14]	<1e-4			

G Null layouts in full

G.0.0.1 The grid snap exclusion, in full.

Excluded by the control the pre-registration named and by a stronger one it did not. The detected pitch is one unit in Reactome and BPMN (67% and 63% integer coordinates) and none in WikiPathways (17%, against 0.37 of boxes exactly aligned in x). Displacing every box by up to half a pitch leaves the held fraction at 0.500, 0.206 and 0.889 against 0.520, 0.207 and 0.912. That test cannot fail at this pitch, and the reason is the instrument: T_A is $\min_j \min(|\Delta x|, |\Delta y|)$, so a box offset by ϵ from its nearest agreement is released only when a move of radius d lowers the term, and moving d toward the agreement lands at $d - \epsilon$, which is worse whenever $\epsilon < d/2$. Half a pitch is 0.5 units against a $d/2$ of 8, 8 and 13. The control that does discriminate is a layout snapped to the same grid, with the same boxes and the same bounding box and no author: it holds 0.065, 0.075 and 0.067 (Figure 4). Displacing the hand layouts past $d/2$ takes them to that same level (0.097, 0.095 and 0.200 at a half-width of 15, 0.065, 0.061 and 0.062 at 120). The grid is available to the random layout and does not produce the effect. Two additions after review: a random layout snapped to the plausible working pitch of five units holds 0.076, the same 0.07 as at pitch one, so the null is not flattered by the finest pitch; and the jittered hand layouts begin falling well below $d/2$ (0.59 on BPMN at a half-width of 5), so the corner argument bounds only the declared half-pitch test, and the caption of Figure 3 says so.

XX:22 What Holds a Hand-Drawn Diagram?

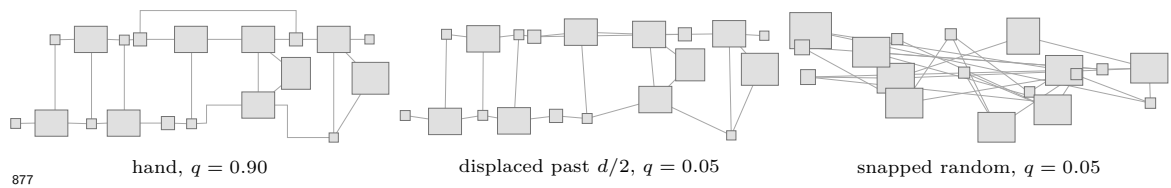


Figure 4 The grid alternative of Section 7: the grid alone does not produce the alignment hold. One BPMN model, 19 boxes; the corpus medians are 0.91, 0.20 and 0.07. Left, the drawing as its author left it. Centre, every box displaced by up to 15 units, just past the $d/2$ below which the directional test cannot resolve a misalignment. Right, every box placed uniformly at random in the same bounding box and snapped to the same grid: the grid is available and the author is not. Box sizes are unchanged throughout. The hand panel draws the routes the file stores; the two control panels draw chords, their boxes no longer being where the routes ran.

G.0.0.2 Null layouts.

Review asked for the chance line under every term, not only alignment. Uniform random layouts of the same graphs, chord geometry, one draw per diagram, seeds from the diagram ids: alignment 0.06 to 0.07 against hand 0.52, 0.21 and 0.91; node-edge 0.11 to 0.12 against 0.76 to 0.87; flow 0.26 to 0.41 against 0.59 to 0.85; orthogonality 0.00 against 0.05 to 0.40. Length and stress hold 0.00 in the random layouts too: no layout of any provenance is stationary under a distance term at this radius except a tool's converged output, so the distance result separates optimisers from everyone, not humans from tools. Crossings and overlap hold 0.58 to 0.67 of random boxes, the flat base rate the active column already prices. Under a jitter of half a grid cell either way the alignment hold keeps 0.76, 0.44 and 0.19 of its three corpora while orthogonality falls from 0.40 to 0.18 and 0.16 to 0.06: orthogonality, not alignment, is the tie artefact. Axis-only moves, which preserve every perpendicular alignment tie, still improve length and stress on the median diagram of every corpus (q 0.00, means 0.005 to 0.016), and a weight of 0.001 on length collapses the held fraction of $C+O+A_1$ from 0.89, 0.51 and 0.22 to 0.13, 0.11 and 0.07: the distance terms are unpriced even inside the alignment-preserving move set, and no small weight on them survives beside alignment. The constrained model of the layout literature, stress minimised subject to alignment constraints [39, 40, 41], fails under its own move set: restricted to the directions that do not raise A_1 , 0.87 to 0.90 of the median diagram's boxes still have a stress-lowering move, and the boxes alignment pins entirely, no direction admissible, are 0.02 to 0.05 (`results/constrained.py`). The generator and every number: `results/nulls.py`.

H The other alternatives

The rest, each with the test that excluded it and the numbers it turned on:

alternative	test	result
Reference length	L at the median edge length and at $1/\sqrt{N}$	hand held on 0.00 to 0.06 of boxes under length and stress, as under the fit; only the control moves
Straight segments where the person drew polylines	primary tables on the stored routes, chords beside them	no verdict moves
A constrained energy whose constraints are absent	the flow term in the tables; the BPMN lanes decomposed	flow holds 0.59 to 0.85 of hand-placed boxes, so reading direction is a criterion partly satisfied and not the missing constraint; 43% of held BPMN boxes are held by an alignment no horizontal lane can impose, and the lanes of the rest leave a median 6.3 box heights of room. Containment as a hard constraint was not added
The radius	$d \in \{0.005, 0.01, 0.02, 0.05\}$; at $d = 0.02$ the move is about one grid step	the hand verdicts are the same at every d . The table at d in units of box width and of median edge length was not run
Selection: near-trees make overlap and crossings easy	q against m/n over diagrams (<code>sparsity.py</code>)	crossings do track density (Spearman ρ $-0.37, -0.34, -0.11$ by hand and $-0.48, -0.30, -0.41$ for <code>dot</code>), these being near-trees at m/n 1.06, 1.04 and 1.11 against 1.30 in the Rome benchmark [21], so “crossings hold every box” is partly the trees’ gift, given to every layout of the same graphs alike; crossings are what the theory is about [1], and on these diagrams there are few to remove. Overlap does not ($+0.02, -0.06, +0.12$), and length and stress are 0.00 in the sparse and the dense half both. No hand-tool difference is explained
The tools are the wrong tools	ELK layered 0.12.0, the engine the BPMN and KIELER editors run, as a fourth control	its profile is <code>dot</code> ’s: length and stress 0.00, flow 1.00 at value 0.000, alignment 0.42 to 0.62 where <code>dot</code> reaches 0.89 to 1.00

I The pre-registration’s scorecard

An earlier version of this study took 147 BPMN models without the connector filter, two fifths of them read as bipartite graphs.

The registered $d =$ one grid pitch (`results/pitch.md`), 0.0008 to 0.0017 of the width per diagram: hand length and stress hold 0.00 with 0.99 to 1.00 of diagrams exactly there, A1 keeps its gap (0.11 and 0.83 against `neato`’s 0.00 and 0.04), and `neato`’s stress at 0.76 and 0.80 extends the radius sweep’s left edge. WikiPathways has no detected pitch, out of scope as registered.

I.0.0.1 The pre-registration’s scorecard.

H1, crossings and overlap hold the hand layout above 0.9 and above `neato`: holds, 1.00 against 0.41 to 0.76. H2, length alone holds under 0.2 and below `neato`: holds, 0.00 against 0.24 to 0.42. H3, no positive length weight repairs $C+O+L$: holds, fitted weight 0.000 everywhere and q 0.008 at any weight past 0.001. H4, alignment holds more boxes than orthogonality or node-edge in every corpus: **refuted**. Node-edge holds 0.88, 1.00 and 1.00 of hand-placed boxes against alignment’s 0.52, 0.21 and 0.91. Node-edge is priced in the standard energy and alignment is not, so the omitted-term conclusion stands, but the registered ranking was wrong and the recorded prediction with it. H5, exploratory, stress holds the hand layout less than `neato`’s: holds, 0.00 against 1.00. The registered fourth

XX:24 What Holds a Hand-Drawn Diagram?

corpus, HOLA's formative drawings, was dropped undeclared from the previous draft; it is restored above. The deduplication pass is run in Section 4 and the 41–100 band above. Registered and still not run: the caps in pitch units.

J Deduplication and drawers

194 of the 248 Reactome graphs and 120 of the 300 BPMN graphs in the band are the component of a larger diagram; no WikiPathways pathway in the band is.

The 136 drawings are 17 participants crossed with 8 graphs, so the interval is clustered: resampling participants (2000 draws) puts the alignment median at [0.31, 0.57], and the per-participant medians run 0.14 to 0.96. Across the four corpora the seven terms every corpus measures rank in the same order, Kendall's $W = 0.99$.

The registered deduplication pass (`results/dedup.py`): by exact coordinate multiset, one Reactome pair is a single drawing stored under two ids; by the structural key (n , m , degree sequence, box sizes), one further Reactome pair and one WikiPathways pair; the 300 BPMN models collide under neither key, so the feared resubmission duplicates are absent from this sample. Removing every extra member moves no headline median. Clustering by drawer: the 305 WikiPathways pathways carry 153 first authors, the most prolific 9% of the corpus; the A1 median's interval clustered by first author is [0.48, 0.55] against [0.48, 0.54] over diagrams, and the twenty drawers of three or more pathways hold medians 0.19 to 0.85 (`results/drawers.py`). Reactome curators and BPMN authors are not recorded in the archives as parsed.

K Reproducibility

Parsers, the energy, the test, the descent, the fitting scripts, the parsed corpora and every table's script are in the repository under `example/diagrams`; the main tables rebuild by `results/run_all.sh`, and every sensitivity of the appendix ships its script beside its committed output. `results/manifest.txt` records commit, platform, tool versions, seeds and the SHA-256 of the archives and the corpora as measured. The tracked `data/bpmn_population_a1.csv` rebuilds the abstract's 0.87 from a clone; the repository carries derived coordinates and ids, never archive content, and the two biological corpora are CC-licensed exports.

L The reinsertion benchmark

The robustness of the rail gain (`results/reinsert.md`): another seed moves it by at most 0.003 and twice the budget by 0.005; the matched-budget uniform-sampling control keeps it (+0.030, +0.020 and +0.036); at the tighter tolerance of 0.0025 it grows on BPMN to +0.085, so the placements are exactly on the rail, not near it; the diagram-clustered 95% intervals are [+0.032, +0.052], [+0.010, +0.026] and [+0.052, +0.074]. Given the neighbourhood, the term picks the person's rail: among placements within 0.05 of the true position under both energies, the rail rate rises from 0.41, 0.35 and 0.53 to 0.55, 0.46 and 0.67 (McNemar $z = 3.9$ to 7.7), while the near-stratum sizes barely move, a further sign the neighbourhood is not alignment's doing.

M Related work, tabulated

work	what it did	what it gives, and what it leaves
<i>By having people draw</i>		
van Ham, Rogowitz [16]	73 hand layouts against a force-directed algorithm, half started from its optimum	71 of 73 raise edge-length variance above the optimum; 60% fewer crossings
Dwyer et al. [17]	32 participants laid out one graph; 194 judges ranked the results against three automatic layouts	judges' choices track stress, without a test; its Table 2 has user layouts at 3 to 9 times the force-directed stress
Purchase, Pilcher, Plimmer [25]	participants drew from adjacency lists	crossing removal the strongest aesthetic, alignment to a grid important
Kieffer et al. [13]	17 participants' layouts of 8 small graphs, ranked by 66 judges	gridiness correlates with rank on all 8, the source of HOLA's alignment step; rank tracks stress on 4 of 8
Mooney et al. [11]	preference and shortest-path performance against stress, graphs to 50 nodes	people prefer low stress and perform better as it falls; why the conclusion adds a term and removes none
Mooney et al. [12]	4,890 drawings harvested automatically from 27 editions of these proceedings	measured in Section 6: length and stress 0.00, alignment 0.87. Provenance is not per drawing and edge directions are dropped, so flow and the box terms are not run
<i>By showing people drawings</i>		
Purchase et al. [18]	a Turing test on hand and algorithmic drawings	hand drawings distinguishable, and judged better
Klammler, Mchedlidze, Pak [14]	Nelder-Mead fit of a linear aesthetic combination discriminating good from worsened layouts	no human coordinates enter it; they call human-labelled data a critically important accomplishment still to come
<i>By learning from examples</i>		
Masui [15]	a weighted constraint combination from good and bad layouts, by genetic programming	neither tested whether the examples are minima of the learned energy
Spönemann et al. [26]	weights adjusted from user selections among candidates	as above
Elud [19]	annealing continued from crowd layouts of pathway diagrams, three networks	the closest prior descent from a hand layout; it raised an asserted five-term score and read that as success. The cap, the per-term test, the reference and the controls here are what make it a measurement of the score
Zhang et al. [28]	64,222 pairwise preference labels	the nearest recent work; it learns from preferences, not coordinates
Wang et al. [42]	a graph network drawing arbitrary graphs under a weighted sum of aesthetics	the weights are asserted or tuned, not learned from people; no human coordinates enter it
O'Donovan, Agarwala, Hertzmann [8]	a 122-parameter energy over single-page graphic designs fitted to a handful of human examples by nonlinear inverse optimisation	each example taken as optimal under the unknown parameters: the assumption examined here, and the closest prior instance of it we found